ROLL NO: | | | | | | | | | | |

(To fill the roll no is compulsory)

INVERTIS UNIVERSITY BAREILLY

EVEN SEMESTER EXAMINATION - 2024-25

DIPLOMA (CSE)

Semester: VI

SUBJECT CODE: DCS603

COMPUTER NETWORKS

Time: 03:00 Hrs. **Maximum Marks: 60**

NOTE: THE QUESTION PAPER CONTAINS THREE SECTIONS. ATTEMPT ALL THE SECTIONS.

SECTION – A

NOTE: Attempt all parts of the following questions. (12 Marks)

Q. 1. Define the following terms: **(1×6 = 6 Marks)**

(i) Computer Networks
(ii) Network Topology
(iii) Parity Bit
(iv) IP Address
(v) Cryptography
(vi) HTTPS

Q. 2. Answer the following questions: **(1×6 = 6 Marks)**

(i) Define transmission media.
(ii) What is circuit switching?
(iii) Name the two main types of sliding window protocols.
(iv) What is the function of repeater in a network?
(v) What is data compression?
(vi) What is the difference between symmetric key and asymmetric key cryptography?

SECTION – B

NOTE:Attempt all the questions. **(6×2×2 = 24 Marks)**

Q.3. (i) (a) In mesh topology, how many physical links are required for a network of 8 devices?

OR

(b) Define star topology. Also list its disadvantages.

(ii)(a) How does a single-bit error differ from a burst error?

OR

(b) What is cyclic redundancy check (CRC)?

Q.4. (i)(a) Write an example of a valid IPv6 address.

OR

(b) What causes network congestion? Discuss in detail.

(ii)(a) What is Huffman coding technique?

OR

(b) Define the structure of URL with a suitable example.

SECTION – C

NOTE: Attempt all the questions. **(12×2 = 24 Marks)**

Q. 5.(a) Differentiate between guided and unguided transmission media. Give two examples of each with their uses.

OR

(b) Compare and contrast the OSI and TCP/IP models.

Q. 6. (a) Describe how routing works in a network and explain the difference between static and dynamic routing.

OR

(b) What is a firewall in the context of network security? Discuss how a firewall works to prevent unauthorized access to a network.